

Gebze Technical University
Dep. of Mechanical Engineering

System Dynamics
and Vibrations (ME 331)

Lecture 2: Equivalent Elements
and Harmonic Motion

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Lecture Overview

2.1 Equivalent spring stiffness

2.2 Coordinates and displacement ratios

2.3 Equivalent mass and rotational inertia

2.4 Equivalent viscous damping

2.5 Energy and small-motion modelling

2.6 Harmonic motion and terminology

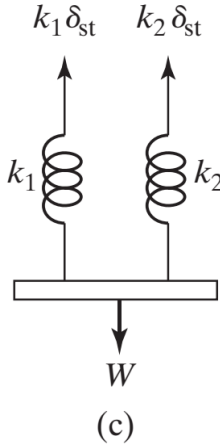
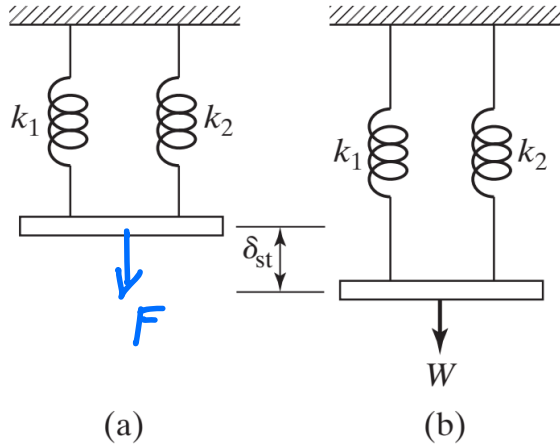
2.7 Displacement, velocity and acceleration

2.8 Phasors and harmonic addition

spring ($F = K\delta$)

$$\delta_1 = \delta_2 = \delta_{\text{tot}}$$

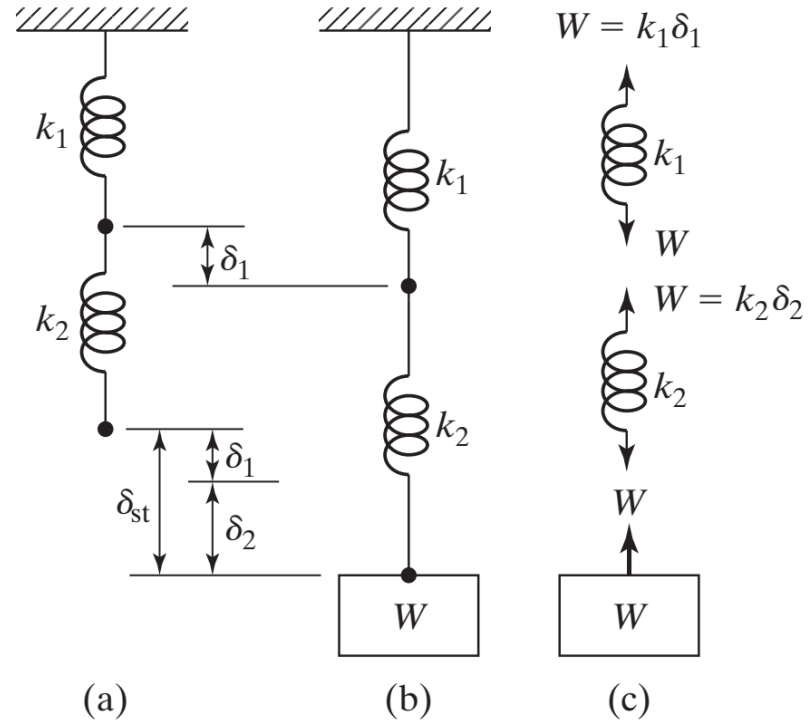
$$F = F_1 + F_2$$



$$K_{\text{eq}} \delta_{\text{tot}} = k_1 \delta_1 + k_2 \delta_2$$

$$\hookrightarrow K_{\text{eq}} = k_1 + k_2$$

$$F_1 = F_2 = W \quad k_1 \delta_1 = k_2 \delta_2$$



$$K_{\text{eq}} \delta_{\text{tot}} = k_1 \delta_1 = k_2 \delta_2$$

$$K_{\text{eq}} (\delta_1 + \delta_2) = k_1 \delta_1 = k_2 \delta_2$$

$$K_{\text{eq}} = \frac{k_1 \delta_1}{\delta_1 + \delta_2} = \frac{k_2 \delta_2}{\delta_1 + \delta_2}$$

$$\delta_1 = \frac{k_2 \delta_2}{k_1} \quad \nearrow$$

$$\frac{1}{K_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2}$$

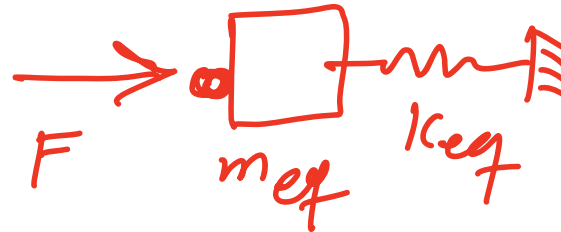
2.1 Equivalent spring stiffness

$$\begin{cases} x_1 = l_1 \sin \theta \\ x_2 = l_2 \sin \theta \end{cases} \rightarrow 1 \text{ DoF.}$$

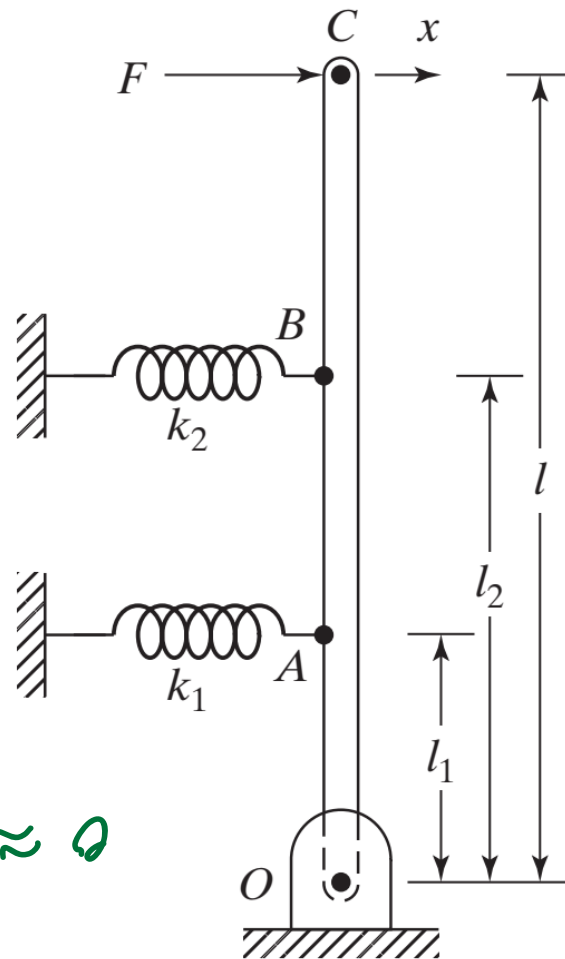
2.2 Coordinates and displacement ratios

$$U = U_g + U_s$$

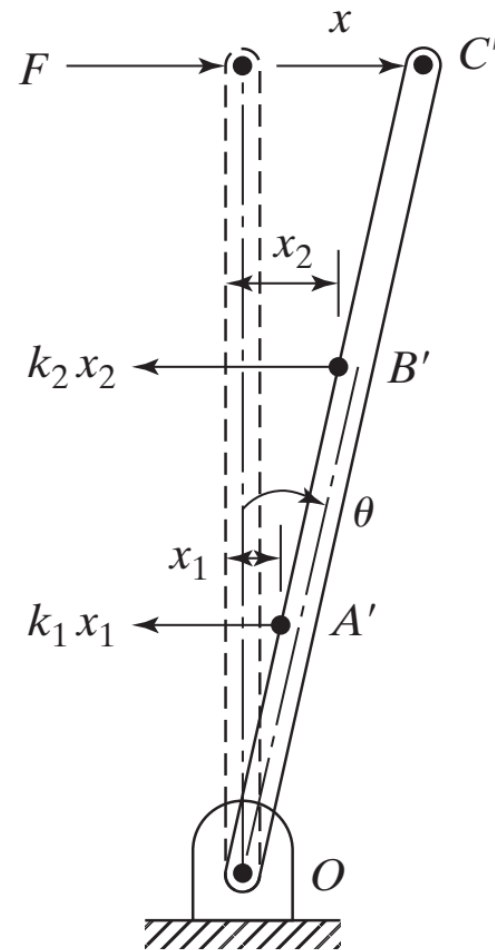
$$\begin{aligned} U_s &= \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 x_2^2 = \frac{1}{2} k_1 (l_1 \theta)^2 + \frac{1}{2} k_2 (l_2 \theta)^2 \\ &= \frac{1}{2} k_1 \left(l_1 \frac{x}{l} \right)^2 + \frac{1}{2} k_2 \left(l_2 \frac{x}{l} \right)^2 = \frac{1}{2} \left[k_1 \frac{l_1^2}{l^2} + k_2 \frac{l_2^2}{l^2} \right] x^2 \end{aligned}$$



$$x = l \theta$$



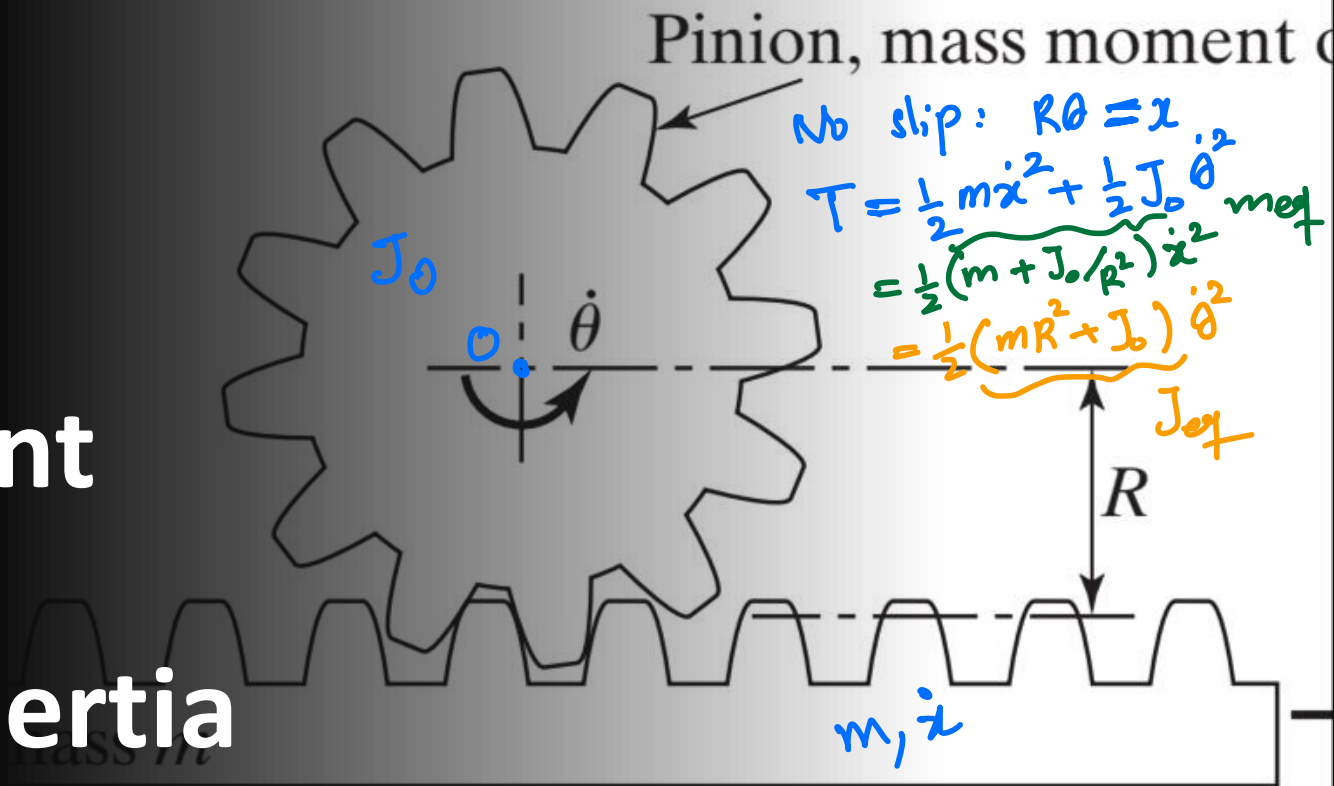
(a)



(b)

$$\sin \theta \approx \theta$$

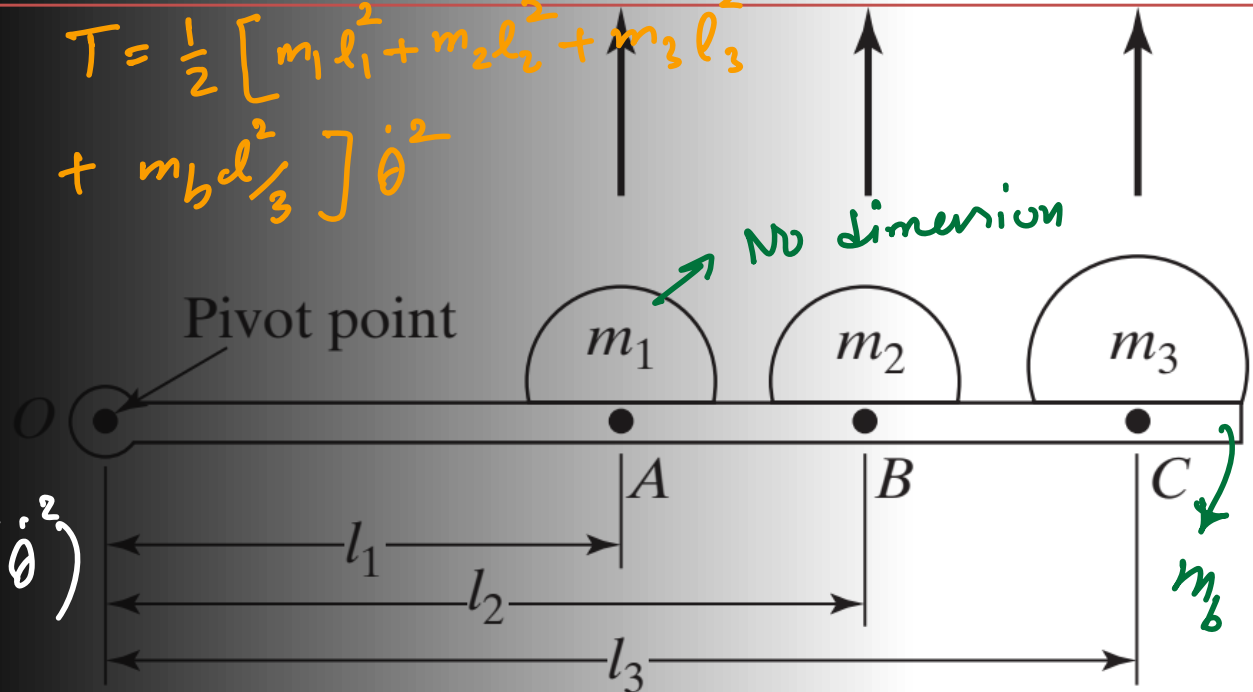
2.3 Equivalent mass and rotational inertia



$$T = \frac{1}{2} m_1 (\dot{l}_1 \dot{\theta})^2 + \frac{1}{2} m_2 (\dot{l}_2 \dot{\theta})^2 + \frac{1}{2} m_3 (\dot{l}_3 \dot{\theta})^2$$

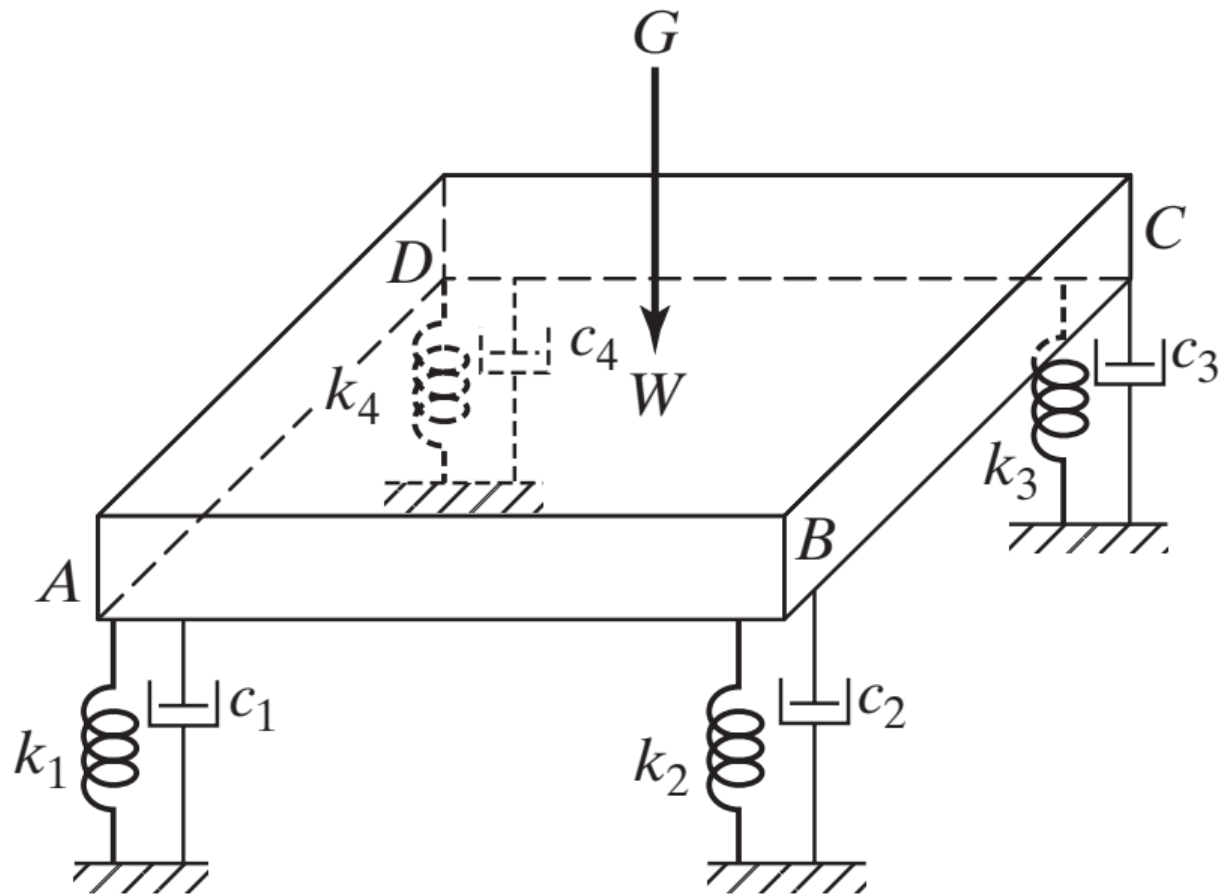
$$\frac{1}{2} I_G \dot{\theta}^2 + \frac{1}{2} m_b v_G^2 + \frac{1}{2} I_G \omega^2 = \frac{1}{2} (m_b \ell_{1/4}^2 \dot{\theta}^2 + \frac{1}{12} m_b \ell^2 \dot{\theta}^2)$$

$$T = \frac{1}{2} [m_1 l_1^2 + m_2 l_2^2 + m_3 l_3^2 + m_b d_{\frac{1}{3}}^2] \dot{\theta}^2$$



2.4 Equivalent viscous damping

$$F = c v_{rel} = c \dot{x}$$



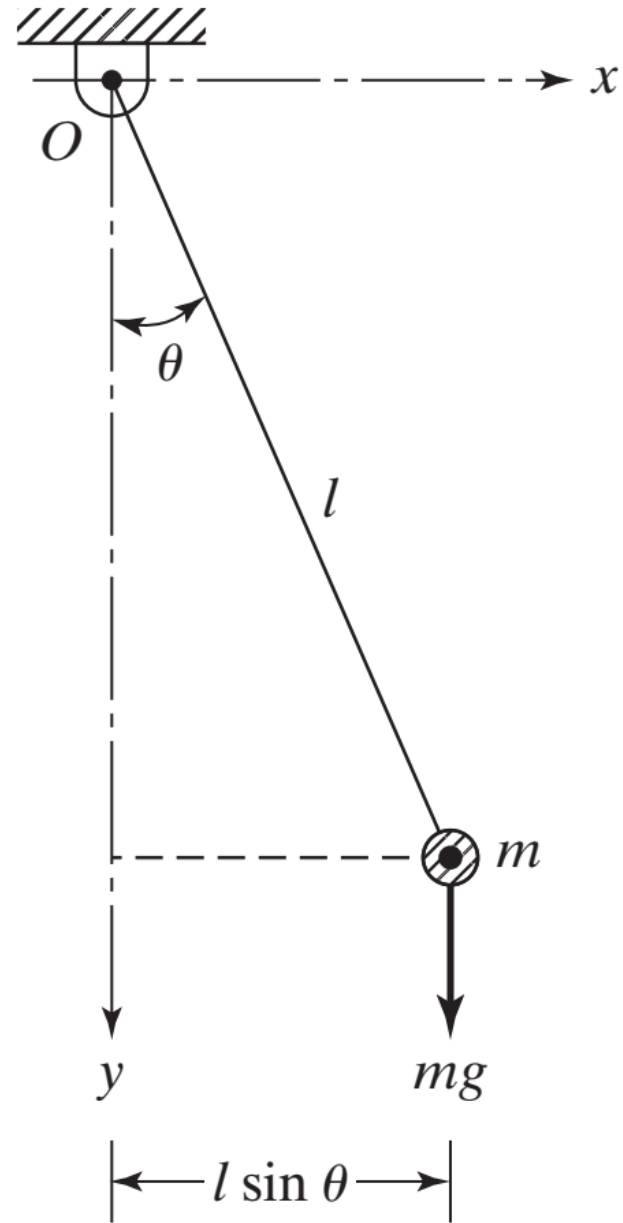
parallel
 $c_{eff} = c_1 + c_2$

series
 $\frac{1}{c_{eff}} = \frac{1}{c_1} + \frac{1}{c_2}$

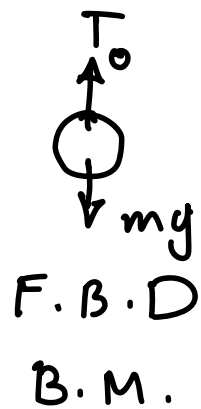
Energy = ?

$$P_{\text{loss}} = \frac{1}{2} c_{eff} \dot{x}^2 \quad \text{loss energy.}$$

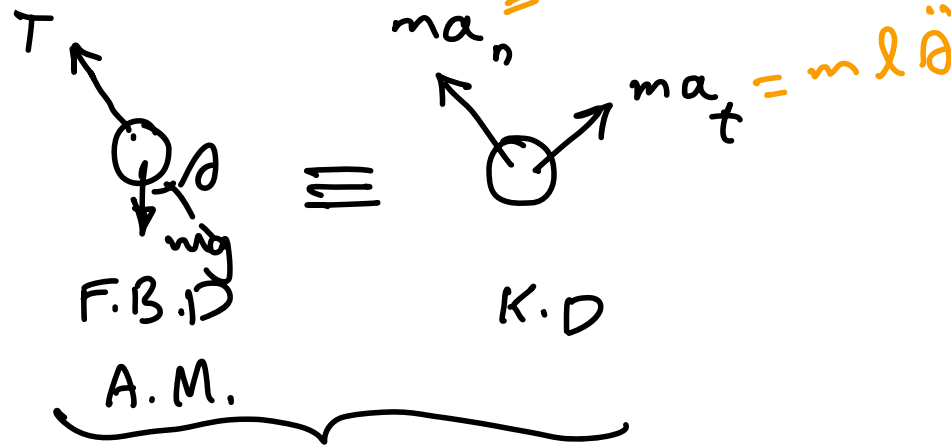
2.5 Energy and small-motion modelling



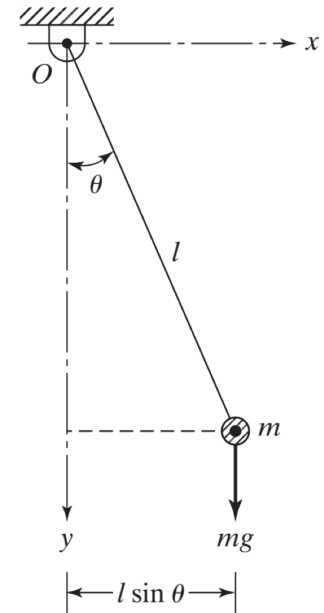
Find governing equations:
method of Newton's law,



$$mg = T_0$$



$$\sum \vec{F} \equiv \sum \vec{f}$$



$$\sin \theta \approx \theta$$

$$t: -mg \sin \theta = ml \ddot{\theta}$$

$$n: T = ml \dot{\theta}^2$$

$$\rightarrow \begin{cases} \ddot{\theta} + \frac{g}{l} \theta = 0 \\ T = ml \dot{\theta}^2 \end{cases}$$

instead
of force:

$$+\circlearrowleft \sum M_O \equiv \sum \mathcal{M}_O$$

$$-mg l \sin \theta = m a_t l = ml^2 \ddot{\theta}$$

$$\rightarrow \ddot{\theta} + \frac{g}{l} \theta = 0$$

method of Energy:

$$T = \frac{1}{2} m v^2 = \frac{1}{2} m (l \dot{\theta})^2$$

$$U = m g l (1 - \cos \theta)$$

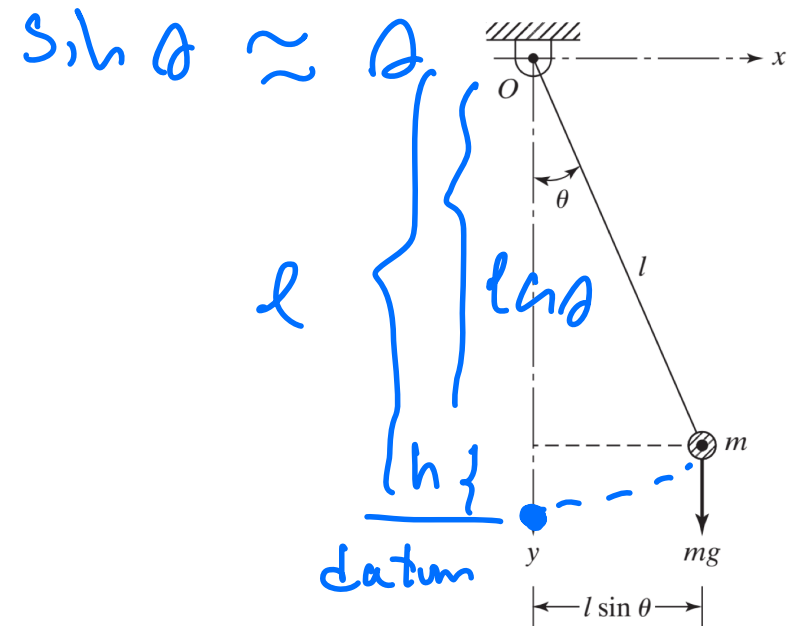
$$T + U = E = \text{constant}$$

$$h = l - l \cos \theta$$

$$\frac{dE}{dt} = 0 \rightarrow m(l^2 \ddot{\theta}) + m g l (\dot{\theta} \sin \theta) = 0$$

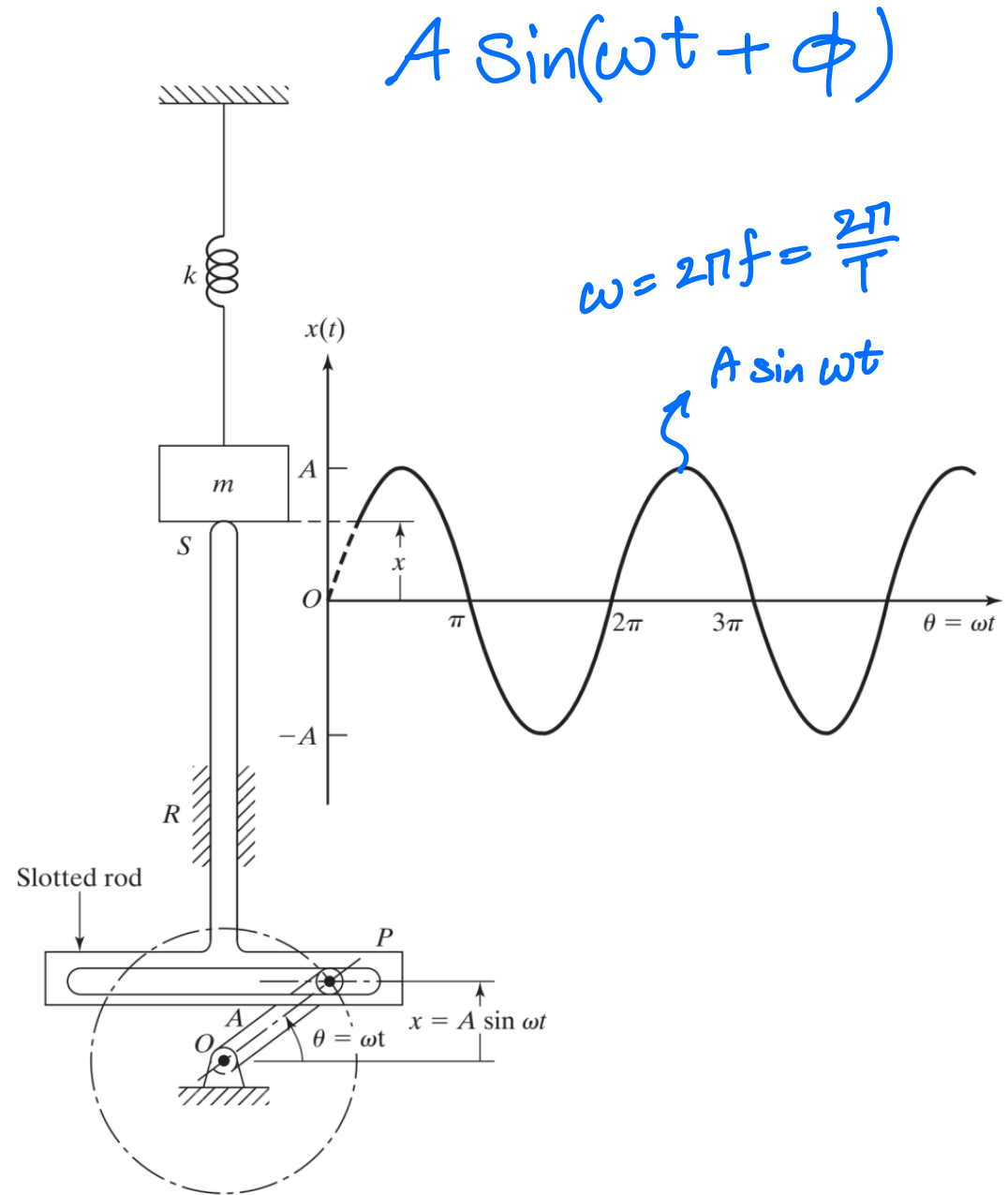
$$\frac{dE}{dt} = \frac{dE}{d\theta} \cdot \frac{d\theta}{dt}$$

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$



2.6 Harmonic motion and terminology

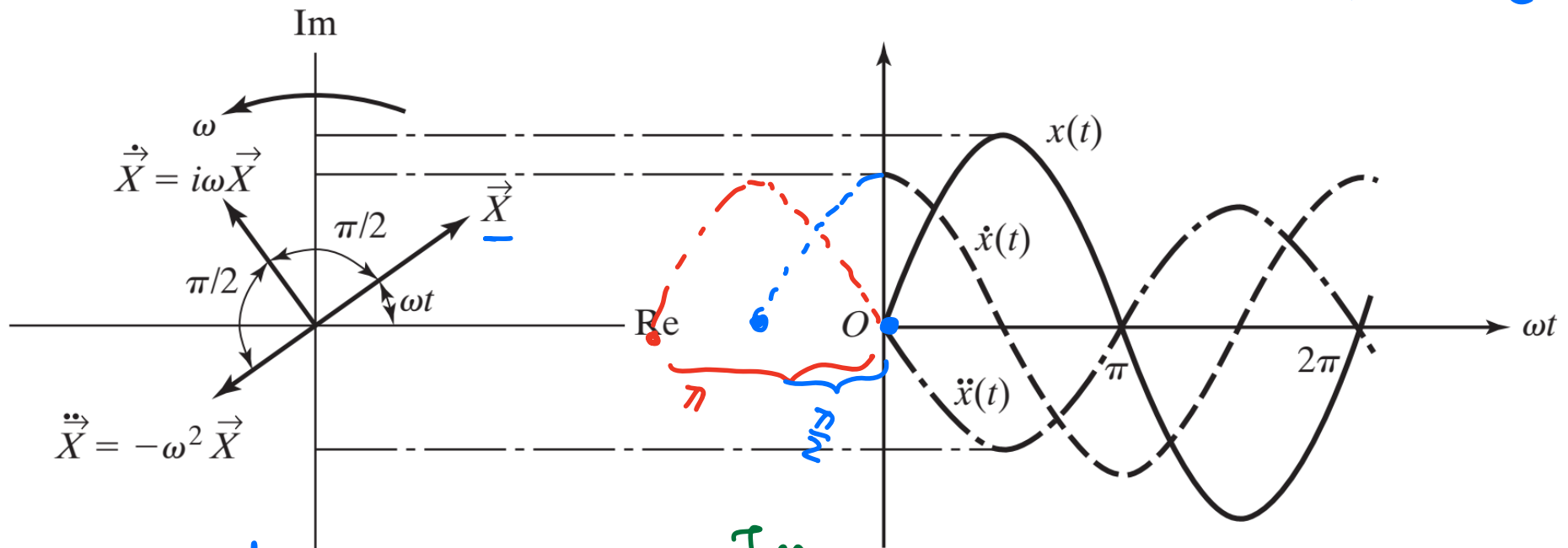
- Amplitude A
- Period T
- Frequency f
- Angular frequency ω
- Phase ϕ



2.7 Displacement, velocity and acceleration

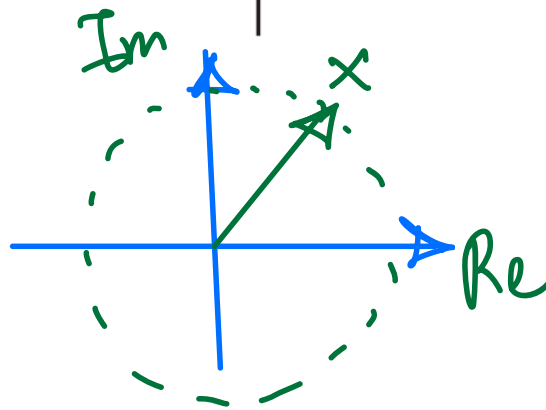
$$\vec{X} : \quad x(t) = X \sin \omega t \rightarrow \dot{x}(t) = X\omega \cos \omega t = X\omega \sin(\omega t - \frac{\pi}{2})$$

$$\ddot{x}(t) = -X\omega^2 \sin \omega t = X\omega^2 \sin(\omega t - \pi)$$



Rotating vectors:

$$\vec{X} = X \cos \omega t + i X \sin \omega t$$



$$\cos(\underbrace{\alpha + \beta}_{\omega t + 2})$$

$$x_2 = 15 \cos \omega t \cos 2 - 15 \sin \omega t \sin 2$$

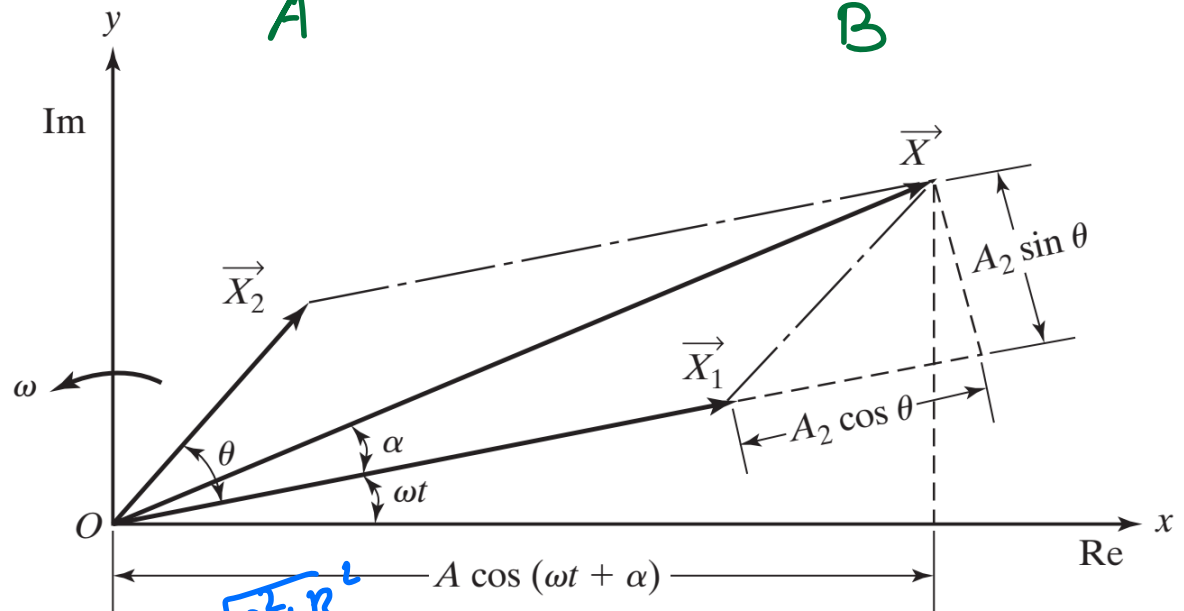
$$x_1 = 10 \cos \omega t$$

$$x_1 + x_2 = \underbrace{(15 \cos 2 + 10)}_A \cos \omega t - \underbrace{(15 \sin 2)}_B \sin \omega t$$

2.8 Phasors and harmonic addition

- $x_1 = 10 \cos(\omega t)$
- $x_2 = 15 \cos(\omega t + 2)$

$$x(t) = x_1(t) + x_2(t)$$



$$\sqrt{A^2 + B^2} \cdot \left(\overset{\cos \gamma}{\frac{A \cos \omega t}{\sqrt{A^2 + B^2}}} - \overset{\sin \gamma}{\frac{B \sin \omega t}{\sqrt{A^2 + B^2}}} \right) = \overset{\sqrt{A^2 + B^2}}{(\cos \gamma \cos \omega t - \sin \gamma \sin \omega t)}$$

$$x(t) = \sqrt{A^2 + B^2} \cos(\omega t + \gamma), \quad \gamma = \tan^{-1} \frac{B}{A}$$

